Luigi Petrucco

Neuroscientist & AI Researcher

luigi.petrucco@gmail.com
vigji.github.io
vigji

From animal behavior to model behavior.

Behavioral and systems neuroscientist (PhD, Max Planck) working in AI engineering, with a growing focus on the behavior of AI models.

AI Safety & Model Behavior

Oct 2026 **ARENA 9.0 — Alignment Research Engineer Accelerator**, *ARENA*, London, UK
Selective in-person programme in alignment research engineering: transformer internals and mechanistic interpretability, reinforcement learning and RLHF, and LLM evaluations.

Jul 2026 **Technical AI Safety course**, *BlueDot Impact*
Intensive course on technical AI safety: alignment, interpretability, and evaluations of frontier models.

2026 **Independent research on LLM behavior**
Persona-conditioned elicitation in the Mistral model family (LessWrong post).

2026 **Red-teaming & adversarial robustness**, *Competitive leaderboards*
Jailbreaking and prompt injection on frontier LLMs and agents: **Top 30** on the JudgmentDay red-teaming arena (AIM Intelligence); **Top 50** in the Gray Swan indirect prompt-injection challenge.

Neuroscience Research

Mar 2022 – Aug 2025 **EMBO Postdoctoral Researcher**, *Istituto Italiano di Tecnologia*, Rovereto, Italy
Neural and computational bases of pursuit behavior in freely moving animals: deep-learning pose estimation, hidden Markov models segmenting behavior into latent states, and links to large-scale electrophysiology. Supervised MSc and PhD students.

Sep 2017 – Apr 2022 **Ph.D. Researcher**, *Max Planck Institute of Neurobiology / TUM*, München, Germany
Discovered a ring-attractor circuit acting as a heading compass in the larval zebrafish brain, combining closed-loop virtual-reality behavior with whole-brain imaging. Designed the closed-loop behavioral paradigms and built the open-source platform to run them (Stytra). Thesis awarded **Best European PhD Thesis in Neuroscience 2022–2024** (FENS-Kavli).

AI Engineering

Mar 2026 – present **AI Unit Lead**, *Afliant S.r.L.*, Rovereto, Italy
Leading the company's AI unit, **supervising a team of 10 people**. Directing the team's evals and improvement of an agentic LLM orchestrator for marketing research.

Sep 2025 – Mar 2026 **Data Scientist & AI Engineer**, *Afliant S.r.L.*, Rovereto, Italy
Developing and evaluating AI solutions for business clients; driving the company's adoption of AI-assisted software development. Synthetic multi-agent LLM populations for marketing research; ML pipelines on Airflow, MLFlow and Databricks.

Sep 2022 – present **AI Engineering Consultant**, *ICOP S.p.A.*, Udine, Italy
Embedded data and AI function within an Italian engineering and construction firm. *Concurrent role.* Designed, evaluated and deployed multimodal VLM pipelines for document matching; leading the rebuild of legacy software with AI coding agents.

Open-source & teaching

Sep 2020 – Apr 2022 **Founder & core developer**, *Brainglobe*
Python ecosystem for computational neuroanatomy. **Received over $400k in funding**; now maintained at the Sainsbury Wellcome Centre, London. Core libraries: brainglobe-atlasapi, brainglobe-space.

Stytra — Python platform for closed-loop behavioral experiments: stimulation, real-time tracking, and virtual reality.

Apr 2022 – present **Lecturer, *Python for Neuroscience***, *Università di Trento*, Trento, Italy
Founded and teach a Python course for PhD students in Cognitive and Brain Sciences.

Education

2017–2022 **Ph.D. in Neuroscience**, *Ludwig-Maximilians-Universität (LMU) / Max Planck*, München, Germany

2014–2016 **M.S. in Neuroscience**, *University of Pisa*, Pisa, Italy, 110/110 cum laude

2011–2014 **B.S. in Biology**, *University of Pisa*, Pisa, Italy, 110/110 cum laude

2011–2016 **Diploma**, *Scuola Normale Superiore*, Pisa, Italy
Highly selective honors program.

Technical skills

Experiments Behavioral paradigms, closed-loop experiments; HMMs of behavior.

ML PyTorch; deep learning for pose estimation and time-series analysis.

LLMs & agents Evals and elicitation, jailbreaks, prompt injection; RAG, MCP servers, multimodal & agentic pipelines.

MLOps Python scientific stack, SQL, Airflow, MLFlow, Databricks, Docker, GitHub Actions.

Languages Italian (native), English (proficient), German (beginner).

Awards & funding

2024 **Best European PhD Thesis in Neuroscience**, FENS-Kavli Network (2022–2024 cycle).

2023 **EMBO Postdoctoral Fellowship** — two-year postdoctoral funding (~€130k).

2022 Invited talk at CoSyNe 2022; abstract selected from the top 3%.

Selected publications & writing

Luigi Petrucco, Samuel Bortolin, Elena Leonelli, Eric Suardi, Elena Zorzi, Daniele Miorandi, and Davide Piazza. Alyas: Evidence-grounded synthetic populations for simulated social and market research. In *Proceedings of EMNLP 2026: System Demonstrations*, 2026. To appear.

Luigi Petrucco. Hannibal Mistral: the Mistral family has a problem with persona-conditioned elicitation. LessWrong, 2026.

Dasol Choi, …, Luigi Petrucco, et al. Judgement Day: An anatomy of successful multimodal attacks on safety-critical AI systems. Manuscript in preparation, 2026.

Luigi Petrucco, Hagar Lavian, You Kure Wu, Fabian Svara, Vilim Štih, and Ruben Portugues. Neural dynamics and architecture of the heading direction circuit in zebrafish. *Nature neuroscience*, 26(5):765–773, 2023.

Vilim Štih, Luigi Petrucco, Andreas M Kist, and Ruben Portugues. Stytra: An open-source, integrated system for stimulation, tracking and closed-loop behavioral experiments. *PLoS computational biology*, 15(4):e1006699, 2019.

Federico Claudi, Luigi Petrucco, Adam Tyson, Tiago Branco, Troy Margrie, and Ruben Portugues. Brainglobe atlas api: a common interface for neuroanatomical atlases. *Journal of Open Source Software*, 5(54):2668–2668, 2020.